

VERIFIED EXECUTABLE LINEAR ALGEBRA

DenseMatrix makes Lean matrix algorithms run like data.

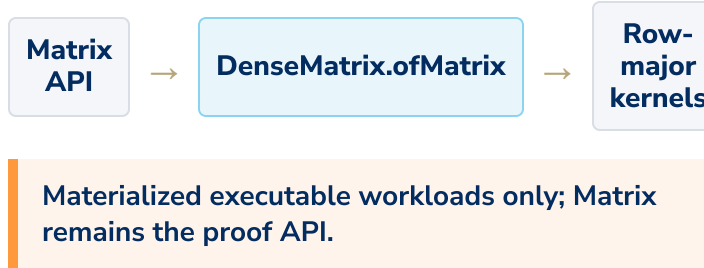
Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly Matrix theorems.

TEAM
Annis Wu · Joseph Qian · Junye Ji
· Veer Shukla
Mentor: Dhruv Bhatia



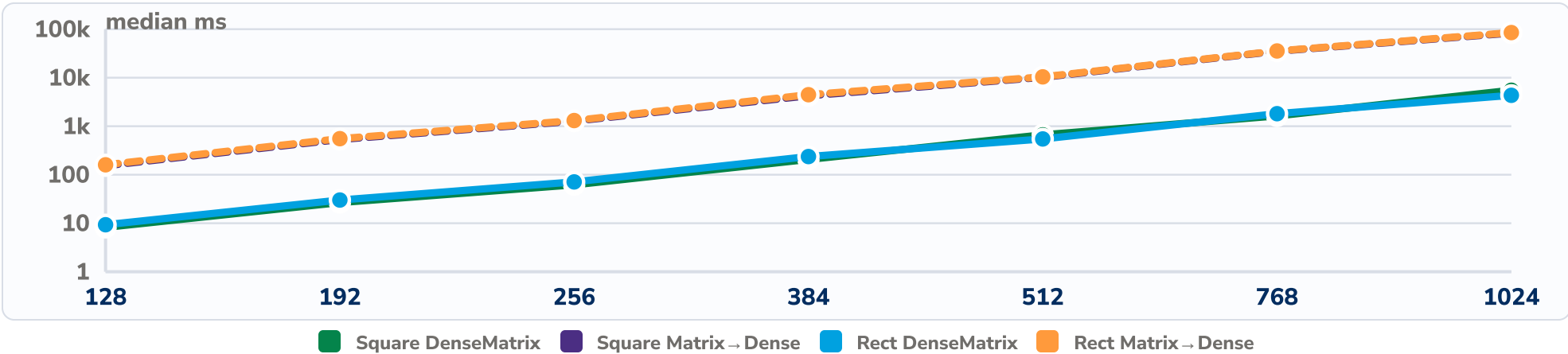
Why DenseMatrix

Row-major arrays let kernels work over contiguous data instead of re-evaluating pointwise Matrix functions.



Multiplication Scaling

DenseMatrix vs Matrix → Dense, multiplication median ms.



Proof Connection

- DenseMatrix.toMatrix_add
- DenseMatrix.toMatrix_mul
- rowEchelonForm_toMatrix
- luDet_eq_det
- gaussDet_eq_det

Theorems connect executable kernels back to the Matrix API.

Representation Tradeoff

Median ms for pointwise primitives at n=1024.

OPERATION	DENSE	MATRIX	MATRIX → DENSE	FASTEST
Add	54.0	45.1	47.5	Matrix
Scalar	53.2	30.3	35.3	Matrix
Transpose	55.3	43.5	46.8	Matrix

Benchmark Scope

2247 RECORDS	328 MATERIALIZED CASES	2048 MAX ROWS
Protocol 30 repeats · 5 warmups		
Sizes 16, 32, 64, 96, 128, 192, 256		
Stress 384, 512, 768, 1024		
Machine pinned CPU core · archived environment		

Algorithm Kernels

Representative rational kernels at n=128.

REF	16,292 ms
RREF	32,394 ms
LU factor	17,090 ms
Gauss det	16,234 ms
LU det	16,933 ms